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Quiz String Matching Part 1

Given an array `text[]`, of size n ($0, 1, \dots, n-1$), of characters

and a pattern of size m ($0, 1, \dots, m-1$).

1) What could a string matching algorithm be used to do? *finding text within a document*

2) Give pseudocode to determine all the occurrences of pattern within text.

```
n = text.size, m = pattern.size
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < m; j++)
    {
        if (text[i+j] == pattern[j] && i+m < n)
            Output "Match found at index " + i;
        else
            break;
    }
}
```

3) What is the running time of your code in terms of n and m ?

$$(n-m+1)m$$